

Recorded Technical Walkthrough

Presenter script · ZW-BioBank v1.0 · Track 1: Data (T1)

Miti AI Consortium · POTRAZ AI for Impact Challenge 2026

Field	Detail
Target runtime	9 minutes (5 required proofs + opening + close)
Presenter	Winston J Mambongo — Presenter & Data Engineer
Lead Innovator	Mutsa M Mutepfa
Primary steward	SIRDC — institutional custody · Miti AI Consortium — technical maintainer
Format	Screen recording with voice-over. Terminal and editor only — no slides.
Repository	github.com/winstonjthinker/MitiAI_Dataset

How to use this script

Every segment gives you three things: **ON SCREEN** is what to display or type, **SAY** is the narration written to be read aloud at a natural pace, and **WHAT THE ADJUDICATOR SEES** is the output that must appear for the proof to land. Read the narration in your own voice — the wording is timed, not sacred. What is fixed is the evidence: every number spoken here is reproducible by running the commands shown.

The rule that carries the most weight is Proof 4. The guide asks for a rule that passes, warns *and* blocks, plus a failed example and its correction path. Segment 5 shows all three states on screen. Do not rush it.

Before you hit record

#	Pre-flight step
1	Restore a clean package state so the numbers on screen match this script: <code>python scripts/validate.py</code> then <code>python scripts/build_manifest.py</code>
2	Confirm the seal is intact: <code>python scripts/build_manifest.py --verify</code> must print OK — package intact .
3	Terminal font at 16–18pt minimum. Adjudicators may watch on a laptop; 11pt is unreadable after compression.
4	Maximise the terminal and clear scrollback before each command so output starts at the top.
5	Close Slack, mail and notification pop-ups. A notification banner over a governance refusal is an avoidable loss of credibility.
6	Open the repository folder in your editor in a second window, ready for the schema segment.
7	Do a silent dry run of the full command sequence once. Confirm nothing errors before you narrate over it.

Segment map

Clock	Segment	Proof
0:00 – 0:45	Opening — what this dataset is and who it serves	Context
0:45 – 2:00	Load the data	Proof 1
2:00 – 3:30	Trace provenance	Proof 2

Clock	Segment	Proof
3:30 – 5:00	Explain the structure	Proof 3
5:00 – 7:15	Run a quality check — pass, warn and block	Proof 4
7:15 – 8:30	Show governance	Proof 5
8:30 – 9:00	Close — handover and sustainability	Close

The script

0:00 – 0:45 OPENING

ON SCREEN

Terminal open at the repository root, nothing running yet.

```
$ ls
```

SAY

Zimbabwe holds one of the richest medicinal plant traditions in southern Africa, and almost none of it is in a form an AI model can learn from. ZW-BioBank changes that.

This is a sovereign, controlled-access, multi-modal biological dataset of Zimbabwe. It links four streams on one physical sample — environmental context, genomic sequence, metabolomic profile, and the traditional knowledge that told us which plant to collect in the first place.

It supports a bioactivity prediction task: given a plant and its chemistry, predict therapeutic potential. The beneficiaries are Zimbabwean researchers who currently export samples to analyse them, and the indigenous communities whose knowledge directed every collection — who are guaranteed benefit-sharing when a discovery is commercialised.

I am going to prove five things in the next eight minutes. That the data loads. That every record traces to a consenting holder. That the structure is documented. That a quality rule can block a release. And that the governance is enforced by the database, not by a promise.

0:45 – 2:00 PROOF 1 — LOAD THE DATA

Minimum proof required: Open a representative record or file in a standard tool and confirm it is machine-readable.

ON SCREEN

```
$ cd mitiai_zw_biobank_v1.0_ai4i_data
```

```
$ python scripts/machine_loading_test.py
```

Then open **processed/heritage_knowledge_v1.csv** in Excel or your editor and scroll one screen.

SAY

No custom parser, no cleaning step. This script loads every processed table using the Python standard library, pandas, SQLite and a QGIS-compatible read.

Three hundred and seventeen records across five tables. One hundred and twenty environmental samples, ninety metabolomic profiles, fifty-five genomic sequences, fifty-two heritage knowledge records, and the QC log the validator writes.

Plain UTF-8 CSV. No merged cells, no multi-row headers, no embedded newlines. Coordinates are WGS 84 decimal degrees, so this drops straight into QGIS as a delimited-text layer. Every CSV also ships as Excel for reviewers who prefer it.

This is a real record. Tsangamidzi — African wild ginger — collected in Chimanimani, rhizome used for cough and asthma, mapped to ICD-11 code CA23, with the Shona original preserved alongside the English translation and marked human-verified.

WHAT THE ADJUDICATOR SEES

RESULT: every file loaded into Python, SQL and pandas with no manual editing.

2:00 – 3:30 PROOF 2 — TRACE PROVENANCE

Minimum proof required: Show where the record came from, who controls the source and how collection and access are managed.

ON SCREEN

Scroll to the provenance chain section already printed by the loading test.

Then open **`governance/source_register.csv`** beside it.

SAY

Machine-readable is not the same as trustworthy. Here is where each record came from.

The database exposes a view called `v_provenance_chain`. It walks backwards from a detected compound, to the physical sample it came from, to the traditional knowledge record that directed the collection, to the holder who shared that knowledge, to their community council's endorsement, to the signed benefit-sharing agreement. Eighty-eight compounds in this pilot resolve all the way to a named holder.

Take this row. Muranga — the pepper-bark tree — bark used for fever and malaria. It links to holder pseudonym HLD-LV-004, council endorsement, and benefit-sharing agreement BSA-ZW-2026-005. If that compound ever becomes a product, the agreement that pays the community is already attached to the row.

The source register lists seven registered sources, from Eastern Highlands field sites to the SIRDC LC-MS core facility. Each one carries its permission status, the evidence for that permission, its known limitations stated openly, and a named fallback source for when access fails. Montane roads wash out in heavy rain — so Nyanga and Mutare wards substitute for Chimanimani, and herbarium vouchers substitute for field identification.

WHAT THE ADJUDICATOR SEES

compounds resolving to a named holder via `v_provenance_chain`: 88

3:30 – 5:00 PROOF 3 — EXPLAIN THE STRUCTURE

Minimum proof required: Show the schema, relationships, dictionary, metadata and versioning.

ON SCREEN

Editor: **schema/schema.sql**, scroll through the table definitions and the three views.

Then **schema/erd.svg**, then **schema/data_dictionary.csv**, then **metadata/dataset_metadata.json**.

SAY

The structure is documented four ways, for four kinds of reader.

schema.sql is the DDL — tables, primary and foreign keys, CHECK constraints, indexes and three views. This is executable, not descriptive. The ERD shows the same relationships visually: environmental samples are the spine, and genomic, metabolomic and heritage records hang off the physical sample identifier.

The data dictionary defines one hundred and fifty-six fields — name, type, units, allowed values, whether it is required, and its sensitivity classification. Units are explicit. gps_precision_m states positional uncertainty in metres, and location_masked declares per row whether that point has been blurred to a ward centroid.

Identifiers are versioned and human-readable by design. ZW-TK for traditional knowledge, ZW-SMP for a physical sample, ZW-SEQ for a sequence, ZW-MET for a metabolomic profile, each with the year and a serial. Language tags follow ISO 639-1 — Shona, Ndebele and English — and disease targets map to ICD-11, so this joins to clinical data without a bespoke crosswalk.

Five metadata cards describe the dataset and each subset, including the bias notes and the quality summary.

5:00 – 7:15 PROOF 4 — RUN A QUALITY CHECK

Minimum proof required: Demonstrate one rule that passes, warns or blocks release, including a failed example and correction path.

ON SCREEN

```
$ python scripts/validate.py
```

Open **validation/error_log.csv** and highlight record ZW-MET-2026-025.

```
$ python scripts/validate.py --demo
```

SAY

Sixteen rules, eight hundred and eighty-eight checks, across all seven quality dimensions — accuracy, completeness, consistency, integrity, timeliness, uniqueness and validity. Status: BATCH CLEAN. Eight hundred and seventy pass, eighteen warn, nothing fails.

That is the pass state. Here is the warn state. Record ZW-MET-2026-025 is flagged by the `annotation_confidence_honesty` rule. The compound is an unknown sesquiterpenoid dimer at MSI level 4 — that means we detected a feature, we have not identified it. The rule warns that it must never be reported as an identification. The correction path is in the log itself: owner, Pharmaceutical Chemist; action, retain the caveat in any downstream use; status, open. A warning does not block the batch, but it does not disappear either.

Now the block state. This is the part I want to be judged on.

Four governance violations, deliberately attempted. One: recording knowledge a holder classified as sacred — refused by a database CHECK constraint. The proposal says sacred knowledge is never collected; that sentence is a constraint, so there is no field in which such a record could be stored, by accident or otherwise. Two: a knowledge-led sample with no consent reference — refused. Consent is not a field someone remembers to fill in; it is a condition of the row existing. Three: consent dated after the interview it supposedly authorised — refused. The P in prior informed consent is enforced by a constraint, not by a reviewer noticing dates.

Four: a holder withdraws, and a linked sample is left active. The database cannot catch this one, so the validator does. Record ZW-SMP-2026-996 is held as critical, the withdrawal impact view lists every downstream record, and the batch status flips to BATCH HELD. Failing records are not admitted to the bank, and the hold stands until each one is actioned inside our thirty-day commitment.

Four attempted, four refused.

WHAT THE ADJUDICATOR SEES

```
records examined 317 · checks run 888 · PASS 870 WARNING 18 FAIL 0
STATUS: BATCH CLEAN. All records admitted; warnings carry documented caveats.
governance violations attempted in demo: 4 attempted, 4 refused
STATUS: BATCH HELD. Failing records are not admitted to the bank.
```

7:15 – 8:30 PROOF 5 — SHOW GOVERNANCE

Minimum proof required: Explain what is open, controlled or withheld and how the dataset will be handed over and maintained.

ON SCREEN

```
$ python scripts/validate.py
$ python scripts/build_manifest.py
$ python scripts/build_manifest.py --verify
```

Then open `governance/license.txt` and `manifest.csv`.

SAY

First, restore the clean state after the demo, and re-seal the manifest — the validator stamps a run timestamp into the QC log, so those checksums legitimately change on every run.

Verify reports OK: sixty-nine files, every SHA-256 matching. If you clone this repository and run that one command, you can prove the package you hold is byte-identical to the one we sealed. Line-ending translation is disabled in gitattributes precisely so that check survives a clone on Windows.

What is open: the schema, the dictionary, the metadata, the validation code and this representative sample. That is the public tier, and it is what is in the repository. Research access to full record-level data requires a data access agreement with SIRDC. Community-restricted knowledge requires SIRDC plus the community council's concurrence. Commercial use is gated behind an executed benefit-sharing agreement, with no exceptions.

What is withheld entirely: the Holder Identity Register. It is AES-256 encrypted, held in SIRDC custody, and it is not in this package. Every identifier you have seen is a pseudonym. Public coordinates are masked to ward centroids at a stated five-kilometre precision. Sacred knowledge is refused at the database level, so it never enters the bank at all.

Code is MIT. The data is deliberately not under an open data licence — it is sovereign controlled access.

WHAT THE ADJUDICATOR SEES

verify: OK – package intact, every checksum matches

8:30 – 9:00 CLOSE — HANDOVER AND SUSTAINABILITY

ON SCREEN

manifest.csv and governance/budget_refresh_and_handover.csv on screen.

SAY

Handover is a package, not a promise. A README, a manifest with a checksum for every file, machine-readable metadata, and scripts that run on the Python standard library with no network access. SIRDC is the named primary steward with institutional custody; Miti AI Consortium is the technical maintainer. The refresh cycle, budget and access model are in the governance folder.

One thing I want to state plainly: Phase I is a five-hundred-sample pilot specification, not a national inventory. Untargeted LC-MS confidently identifies only a minority of features — most remain putative, and we label them that way rather than overclaiming. Heritage knowledge reflects consenting holders in participating communities, which is a selection bias we monitor with a runnable query rather than a paragraph. Protected-area access is not yet secured.

We would rather be judged on a dataset that states its limits than one that hides them.

We can discover it — from our soil, our forests, and our heritage. Thank you.

Q&A preparation

The session format allows adjudicators to request deeper verification during Q&A. These are the questions most likely to come, with the evidence to open in each case.

Likely question	Answer, and what to open
“Is this real data or synthetic?”	Answer directly. Phase I is a pilot specification package with a representative sample built to the collection protocol — the README states this under Known Limitations. The schema, validation engine, governance constraints and provenance chain are fully real and executable. Open governance/source_register.csv to show the seven sources and their permission status, and governance/progress_log.csv for dated progress.
“Can you show a record failing and then being corrected?”	Run <code>validate.py --demo live</code> . Then open validation/error_log.csv and show the <code>issue_owner</code> , <code>correction_action</code> , <code>corrected_by</code> and <code>resolution_status</code> columns — the correction path is a column set, not a narrative.
“What stops someone re-identifying a knowledge holder?”	Three layers. Identity lives only in the AES-256 encrypted Holder Identity Register in SIRDC custody, never in the package. Coordinates are ward-centroid masked at 5,000 m with <code>location_masked</code> declared per row. Named attribution is opt-in, defaulting to 0. Open governance/anonymization_plan.pdf .
“What is your lawful basis?”	Zimbabwe Data Protection Act (Chapter 12:07), with prior informed consent per holder and community council endorsement per ward. Open governance/dpa_compliance_matrix.csv — each row names its enforcement mechanism, not just the obligation.
“How do you handle a holder who withdraws consent?”	Demonstrated live as demo case 4. The <code>v_withdrawal_impact</code> view lists every downstream record, the validator holds the batch, and the commitment is 30-day propagation. Show consent_log.csv — <code>withdrawal_status</code> and <code>withdrawal_date</code> are first-class fields.
“How is bias measured rather than asserted?”	Open governance/bias_risk_register.csv . Eight risks, each with a runnable monitoring query and a mitigation, including the 30% Ndebele representation floor. Pair with governance/target_population_and_sampling.md .
“Who owns this in three years, and who pays for it?”	SIRDC is the named primary steward with institutional custody; Miti AI Consortium is technical maintainer. Open governance/budget_refresh_and_handover.csv and governance/next_90_days.md .
“Why is the repository public if the data is controlled-access?”	Because the public tier is <i>defined</i> in the licence: schema, dictionary, metadata, validation code and aggregate statistics are open by design. Record-level data, identities and commercial use sit behind the other three tiers. Open governance/license.txt .
“Could an AI team actually train on this?”	Yes for the pipeline and the join logic — the four streams already join on one physical sample, which is the hard part. Volume is the honest gap: Phase I is 500 samples by design, and scale is what Phase II funds.

Recording discipline

Do	Avoid
State a number, then show the command that produces it.	Claiming a figure the adjudicator cannot see on screen.
Let the terminal output finish rendering before speaking over it.	Talking through a scrolling wall of text.
Name the limitation before the adjudicator finds it.	Describing the pilot sample as a national inventory.
Say “putative” for MSI level 3–4 compounds, every time.	Saying “we identified” for a feature that is only detected.

Do	Avoid
Keep Proof 4 slow and deliberate — it is the strongest evidence you have.	Rushing the refusals to save thirty seconds.

Every figure in this script — 317 records, 888 checks, 870 pass, 18 warnings, 0 failures, 88 provenance chains, 4 of 4 refusals, 69 sealed files — was produced by running the commands above against this package. If a number changes because the data changes, re-run and update the narration rather than reading the old figure.